Lecture 35: Transformers



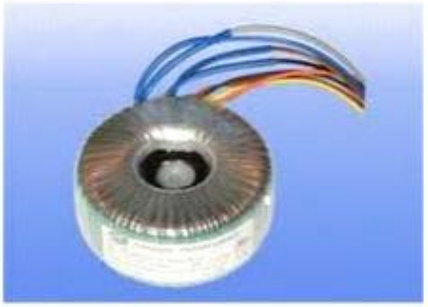






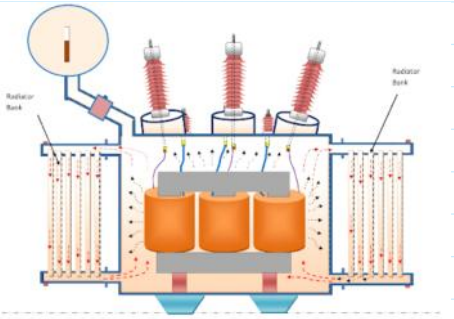








Various Types of Transformers



A. Based on Applications



Power transformer

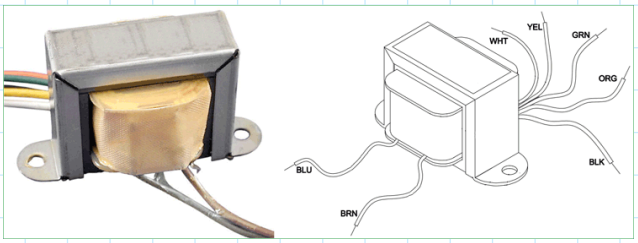


Measurement transformer

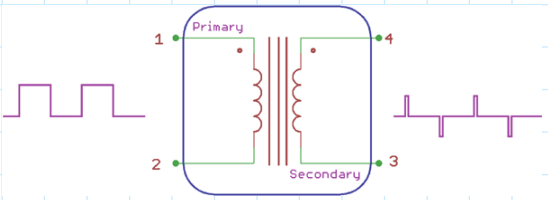
1. Current Trans. (CT)
2. Potential Trans (PT)



Distribution transformer

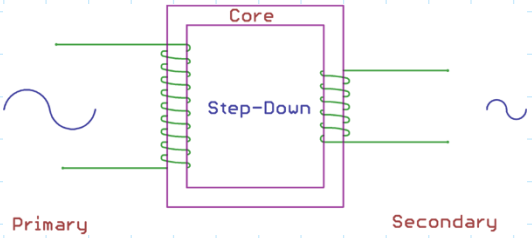


Low voltage transformers

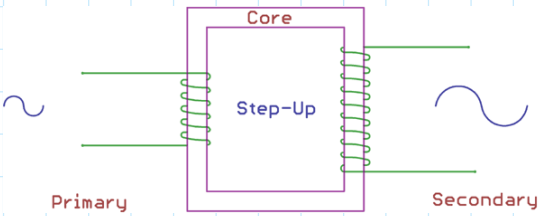


Pulsed transformers

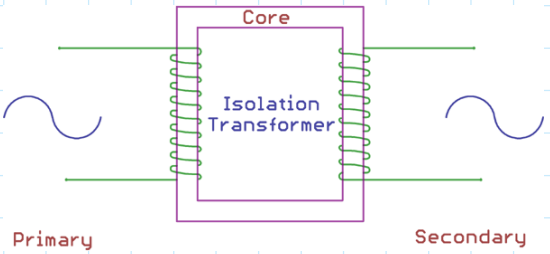
B. Based on Voltage Transformation



Step down transformer

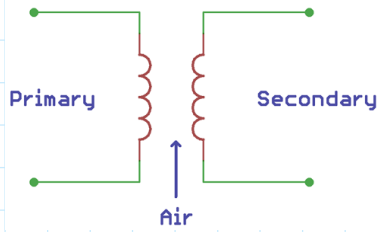


Step up transformer

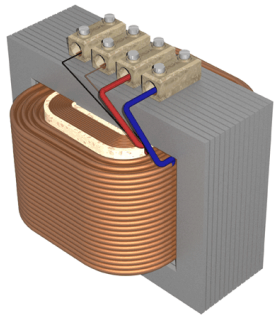


Isolation transformer

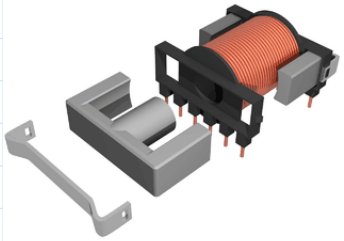
C. Based on Core Materials



Air core transformer



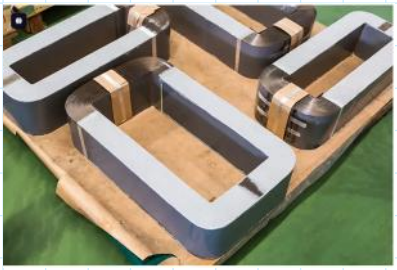
Iron core transformer



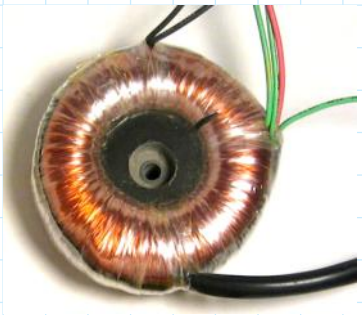
Ferrite core transformer



Core stacking in E or L shape



Amorphous core transformer



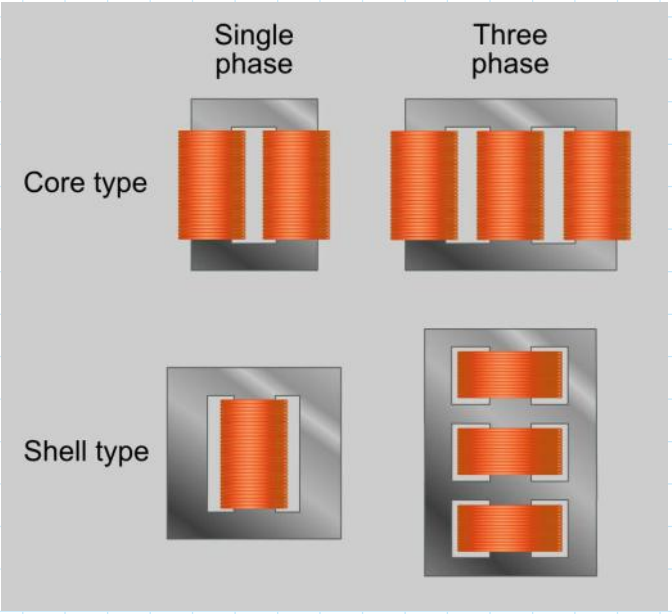
Toroidal core transformer

Comparison of Iron Core, Ferrite Core, Amorphous Core, and Air Core Transformers

Parameter	Air Core Transformer	Iron Core Transformer	Ferrite Core Transformer	Amorphous Core Transformer
Core Material	Air (non-magnetic medium)	Laminated silicon steel	Manganese-zinc or nickel-zinc ferrite (ceramic)	Amorphous metal alloy (non-crystalline)
Electrical Conductivity	None (no eddy current losses)	High (causes eddy current losses)	Very low	Low
Magnetic Permeability	Very low (≈1)	Very high	Moderate	High
Core Losses	Negligible (no core losses)	High (especially at high frequency)	Very low (suitable for high frequency)	Very low (even at power frequency)
Operating Frequency Range	High frequency (radio frequency and wireless power transfer)	50–60 Hz (power applications)	20 kHz–1 MHz (switch-mode power supplies)	50–60 Hz (energy-efficient power transformers)
Efficiency	Low (high leakage reactance)	Moderate	High	Very high
Temperature Stability	Excellent	Good	Limited (loses permeability at high temperature)	Excellent
Size and Weight	Bulky for same power rating	Bulky	Compact	Moderate
Applications	RF transmitters, induction heating, wireless charging	Power transformers, distribution	SMPS, RF circuits, inductors	Energy-efficient distribution transformers
Cost	Low to moderate (depending on design)	Low	Moderate	High

D. Based on Construction

1. Core type (single phase, three-phase)
2. Shell type (single-phase, three-phase)



Source: https://vietnamtransformer.com/

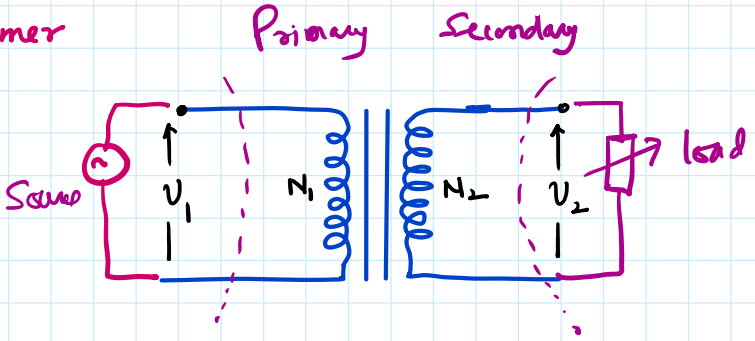
Transformer Animation Video: https://drive.google.com/file/d/1Y8xM5ehdPLw_3aVHTVxt-Z4n1sKx0cGK/view

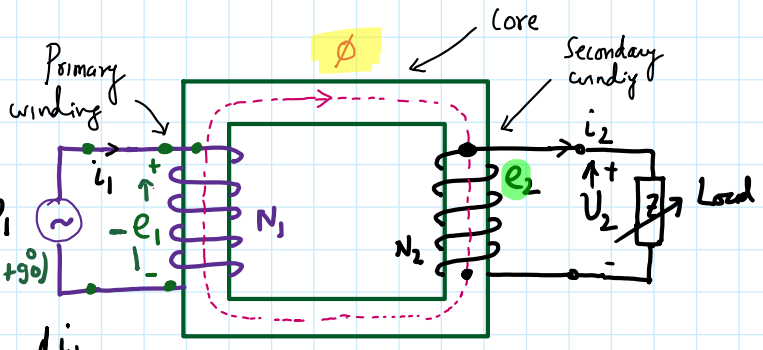
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Principle of Operation of Transformer and its analysis:

Components of Transformer:

- ✓ Magnetic core
  - Soft iron
  - Laminated silicon steel
  - Ferrite core
  - Amorphous core
  - CRGO (Cold reinforced grain oriented steel)
- ✓ Primary winding
- ✓ Secondary winding





$$v_1 = \sqrt{2} V_1 \sin(\omega t + 90^\circ)$$

$$\downarrow$$

$$e_1 = -N_1 \frac{d\phi}{dt} = -L_1 \frac{di_1}{dt}$$

Schematic representation of transformer

For linear relationship between i & ϕ $\Rightarrow$ $L_1 i_1 = N_1 \phi$

Ideal Transformer: Two assumptions

1) No losses in transformer

Ideal Transformer: Two assumptions

1) No losses in transformer
2) Infinite permeability of core material

$V_1 \rightarrow i_1 \rightarrow \phi \;;\; L_1 i_1 = N_1 \phi \Rightarrow \boxed{i_1 = \frac{N_1}{L_1} * \phi}$ (i_ϕ)

$\phi = \phi_{max} \sin \omega t \;;\; \bar{\phi} = \frac{\phi_{max}}{\sqrt{2}} \angle 0^\circ$

$e_1 = -N_1 \frac{d\phi}{dt} = -N_1 * \omega \phi_{max} \cos \omega t$

$\cos \omega t \rightarrow \sin(\frac{\pi}{2} - \omega t) \rightarrow -\sin(\omega t - \pi/2)$

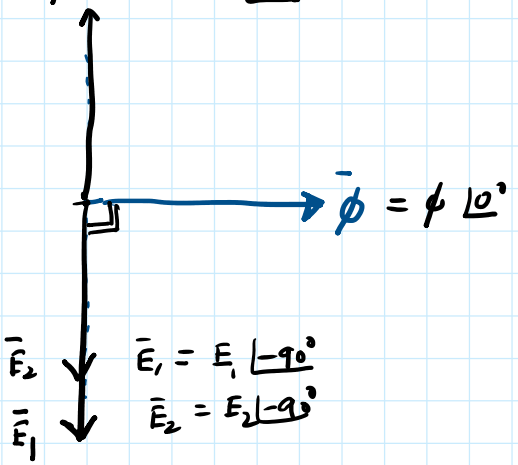
$e_1 = N_1 \omega \phi_{max} \sin(\omega t - \pi/2)$

$\bar{E}_1 = \boxed{\frac{N_1 \omega \phi_{max}}{\sqrt{2}}} \angle -90^\circ$ — (1)

$e_2 = -N_2 \frac{d\phi}{dt} = -N_2 \omega \phi_{max} \cos \omega t = N_2 \omega \phi_{max} \sin(\omega t - \pi/2)$

$\Rightarrow \bar{E}_2 = \boxed{\frac{N_2 \omega \phi_{max}}{\sqrt{2}}} \angle -90^\circ$ — (2)

$\bar{V}_1 = -\bar{E}_1 = V_1 \angle 90^\circ$



Phasor diagram at no load and lossless transformer

$\frac{E_1}{E_2} = \frac{N_1}{N_2} =$ Transformer Ratio $= a$ $=$ Turns Ration

$V_1 \simeq E_1$

$V_2 \simeq E_2$

$\frac{V_1}{V_2} = \frac{E_1}{E_2} = \frac{N_1}{N_2} = a = \frac{I_2}{I_1}$

$V_1 I_1 = V_2 I_2$

$\frac{V_1}{V_2} = \frac{I_2}{I_1} = \frac{N_1}{N_2} = a$

$E_1 = \frac{N_1 \omega \phi_{max}}{\sqrt{2}} = \frac{2\pi f N_1 \phi_{max}}{\sqrt{2}} = \sqrt{2}\pi f N_1 \phi_{max}$

$\boxed{E_1 = 4.44 f N_1 \phi_{max}}$ $\Big\}$ $\frac{E_1}{E_2} = \frac{N_1}{N_2}$

$$E_1 = 4.44 f N_1 \phi_{max}$$

$$E_2 = 4.44 f N_2 \phi_{max}$$

$$\frac{E_1}{E_2} = \frac{N_1}{N_2}$$